Integrating Revenue Management with ERP in Life Sciences Organizations

A Model N Technical White Paper

This white paper is intended as a guide for life sciences information technology executives and managers seeking to understand the implications of implementing RM systems in conjunction with new or existing ERP investments. The white paper briefly describes the high-level requirements and advantages of revenue management in relation to ERP, provides systems of record and integration strategy recommendations, and highlights Model N's successful implementation experiences integrating with SAP R/3.

Introduction: Revenue Management Highlights

Revenue Management (RM) software enables pharmaceutical, biotechnology, and medical technology companies to control revenue leakage and to reduce regulatory risk by enabling sound pricing and contracting practices, and by implementing real-time monitoring, dynamic calculation, and government reporting functions. While the nature of the revenue processes that must be integrated for this picture to work are well understood, IT organizations must approach key architectural and integration decisions as strategic questions because of the implications for business success, risk, and customer satisfaction.

RM aligns pricing, contracting, and settlement lifecycles along key functions:

- Creation and negotiation of profitable offers aligned with customer interests
- Managing complex price and rebate structures as well as other trade arrangements such as administrative and funding fees
- Enforcing accurate pricing on 100% of customer orders
- Policing the risk of violating government price mandates via price tracking, and alerting and filing of specialized government reports
- Reporting and measuring compliance to commercial performance commitments with approvals and lifecycle support
- Validating and processing financial settlements to both commercial and government entities with subsequent integration of payment information to financial systems

What Makes RM a Separate Technical Space from ERP? Why not "just SAP"?

In the overall enterprise framework, RM co-exists with and shares data with customers' existing applications — legacy, custom, or standard ERP systems. Until recently, life sciences companies have tended to invest in extending the standard order-to-cash functionality in ERP or add-on systems. However, the industry and marketplace today is seeing unprecedented customer and regulatory vigilance around government and corporate price compliance. Distribution and payment mechanisms are extremely complicated and data requirements challenging. This has created critical new challenges for IT organizations partnering with business users in the marketing, sales, and finance arenas. RM software can successfully address these challenges today as industry leaders implement a new generation of revenue management systems built from the ground up to integrate with and complement ERP. In doing so, RM software must support several critical requirements:

- **Complex contracts and offer agreements** — Multi-party contracts with complex deal structures are often the order of the day. These require qualifications on multiple pricing points and benefits that are payable on a variety of product groupings. Automatic price monitoring and alerting is required to catch best price violations and non-profitable contracts before they occur.
- **Dynamic relationships** — Active relationships form the basis for all negotiations, commitments, pricing agreements, and payments. Hierarchical customer groups, distributor relationships, membership, eligibility, commitments, and class-of-trade dynamically affect the entire lifecycle of RM.
- **Pricing strategies and price resolution** — Life sciences manufacturers deploy a variety of complex, often parallel pricing mechanisms, such as tiered pricing, quantity-based pricing, capitated pricing, and static/dynamic price lists. At "run time," there is a further need for dynamic price resolution using a variety of strategies when pricing options conflict.
- **Incentive mechanisms and settlements** — A variety of incentives (rebates, admin fees, service fees, and chargebacks or distributor rebates) are very specific to life sciences. These accrue and bucket against specific product groups or products and on a specific time schedule, but frequently settle on a different time schedule against different products.
- **Regulatory compliance** — Apart from the highly specialized government pricing calculations that need to be implemented, RM requires very specific processes, monitors, reports, and tools to support manufacturers that want to do business with government agencies such as FSS, Medicare, Medicaid, or state agencies, and to ensure compliance with SOX or internal controls.

Life sciences companies need to deploy, and *are* successfully deploying, a dedicated RM suite that unifies the entire process of pricing strategy; offer and contract management; rebate and incentive program development; sales and compliance monitoring; and settlement calculations and payments — wrapped around an "always on" regulatory compliance monitoring, alerting, and reporting engine.

Revenue Management Approach: Key Considerations for an Integrated ERP – RM Landscape

There are several critical implementation planning decisions that need to be considered when RM is deployed in an enterprise software environment.

Systems of record: The ERP-RM landscape needs to define the systems of record for data. In most RM implementations, the existing ERP systems are typically the system of record for the customer master, the material/product master, base price lists, direct sales transactions, and invoicing/AP data, to name a few categories. RM solutions are typically the system of record for pricing strategies, contracts, resolved prices, membership and commitment records, chargeback and rebate programs and payments, specialized government price lists, government rebate claims, payments, and disputes. In addition, RM often houses specialized relationship data such as classes of trade, government codes (e.g. 340b/PHS entities, FSS Tracking Entities), plan and group membership, etc. To support the needs of large life sciences companies, a successful ERP-RM landscape suite needs to deploy an integration strategy that handles a variety of data formats across multiple data channels with appropriate monitoring and recovery capabilities.

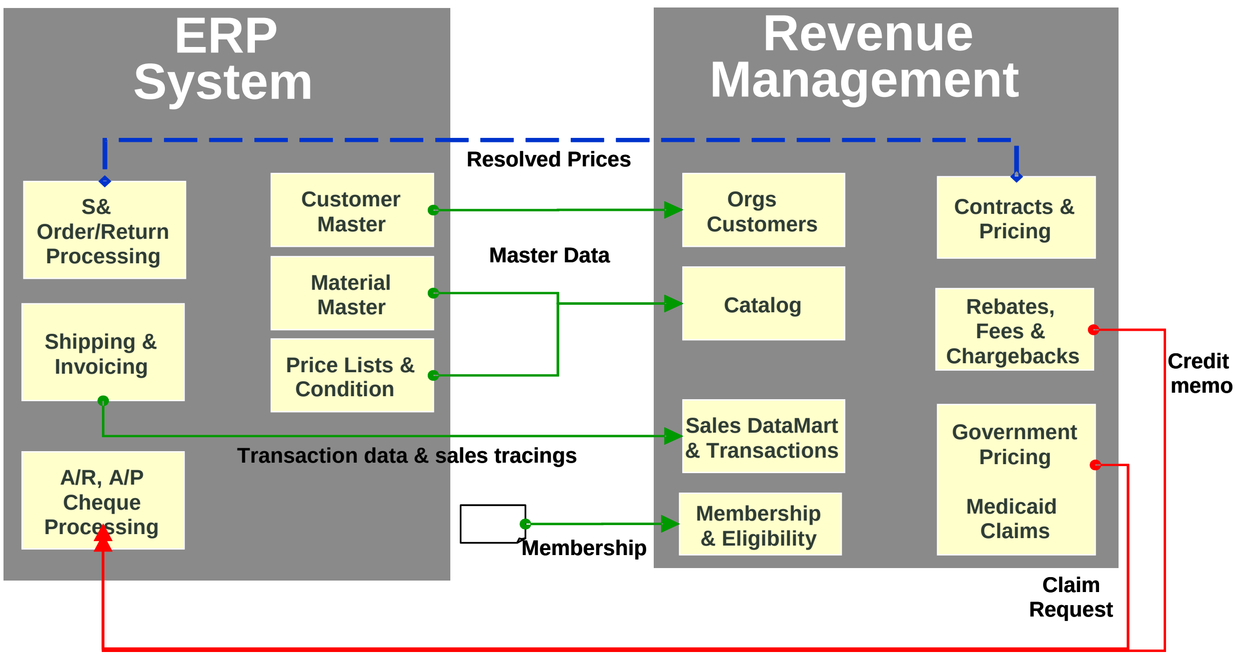
Real-time support for key business processes: ERP and RM systems both require a certain amount of live, real-time financial data. ERP order entry mechanisms need to quote prices based on real-time pricing information. In the life sciences industry, this changes on an ad-hoc basis as membership, eligibility, and contracting parameters fluctuate. This requirement for real-time price resolution implies that the ERP order entry systems (price procedures and advanced pricing modules) are configured to "punch-out" to the RM database for real-time pricing. On the settlement side, many of the related calculations (rebates, chargebacks, and admin fees) are computed and accrued by the RM systems on a scheduled basis. The RM systems initiate appropriate A/P activity in the ERP systems as each of these payments or accruals becomes due.

ERP configuration decisions: RM can drive life sciences specific customizations in the ERP. Even though most RM systems have "out-of-the box" ability to implement and support crucial life sciences customizations, it is frequently best if some customizations are implemented in the ERP system itself. Capabilities such as class-of-trade, customer ID (HIN, DEA, etc.), membership, and specialized pharma product attributes (unit rebate amount) are often best added to the master data in the ERP systems.

Reporting and Analysis: The ERP-RM landscape needs a unified reporting/analysis framework. Data warehouses are frequently deployed to unify reporting across all the payment plans. RM software needs to export controls, data, and intermediate results (or accruals) in a unified fashion to make them continuously available to the larger IT reporting and alerting frameworks.

What Does an Integrated ERP - RM Solution Look Like?

As described above, life sciences RM software manages pricing guidelines and templates, historical and current commercial and government contracts and programs, settlements, membership, and commitment records. Therefore it becomes the system of record for pricing information, rebate and incentive information, contract approval information, government pricing calculations, and a variety of reports and metrics around the revenue management process. The ERP system is the system of record for all customer and organization information, products and materials, order processing, invoice/sales tracing lines, and often other data. In marketplaces where most purchasing is performed off existing pre-negotiated contracts, the pricing (even advance pricing) methodologies of ERP systems are frequently inadequate to reflect the complexities of contract-based and government-aware pricing information typically available in a RM system. Accordingly, the need exists for an integration adapter between ERP and RM solutions.

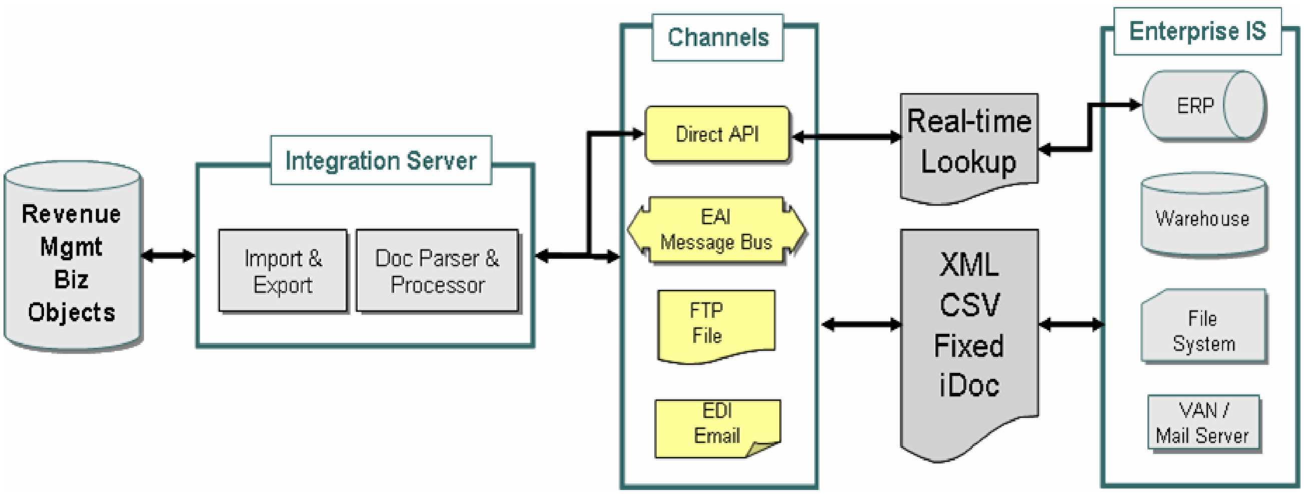


Typical touch points between ERP and RM systems

Legend:
Green = ERP master & transaction data sent to RM
Red = RM financial data sent to ERP A/P
Blue = real-time pricing calls from ERP to RM

RM software needs to incorporate an integration server that provides a framework for executing bi-directional data flow between the RM and ERP systems. Some of the features and benefits of such a server should include:

- A user interface that is integrated with the rest of the RM application for scheduling data flows, managing exceptions, and receiving notifications.
- Importers and exporters for both simple common business objects (such as customer data and base price lists) and complex composed objects (such as structured contracts and government claims disputes)
- Parsers and formatters for multiple document formats (including XML, EDI, CSV, iDoc, and Fixed Column Width).
- Support for multiple channels (such as JMS, IBM WebSphere MQ/WII, WebMethods, SAP/XI, FTP/file, and SMTP), direct API channels (such as LDAP, RFCs, and JCo) and other EAI systems, as required.



The integration server provides a framework for bi-directional data flow between the ERP and RM systems.

Model N and SAP: A Specific ERP – RM Solution (Model N's Adaptors for SAP R/3)

Integration between RM systems and ERP systems usually involves a batch/asynchronous dataflow mechanism for moving large amounts of data, and a real-time API connector for frequent requests of small data amounts.

In the highlighted example, Model N has built specific adaptors for SAP R/3 integration:

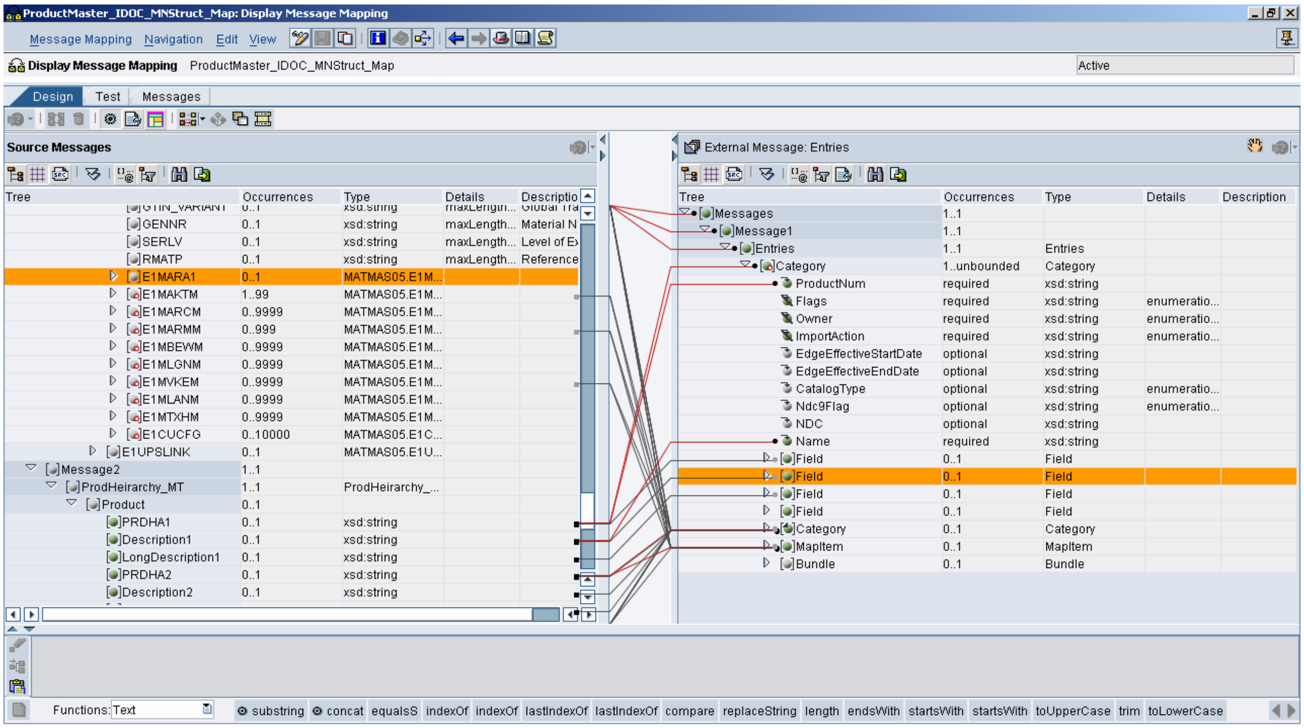
- An **asynchronous data gateway** that exchanges master data, transactional data, and settlement memos between R/3 & Model N's RM suite
- A **real-time connector** that enables price lookup from standard SAP pricing procedures to the Model N Price master

The Model N **asynchronous data gateway** to R/3 transfers:

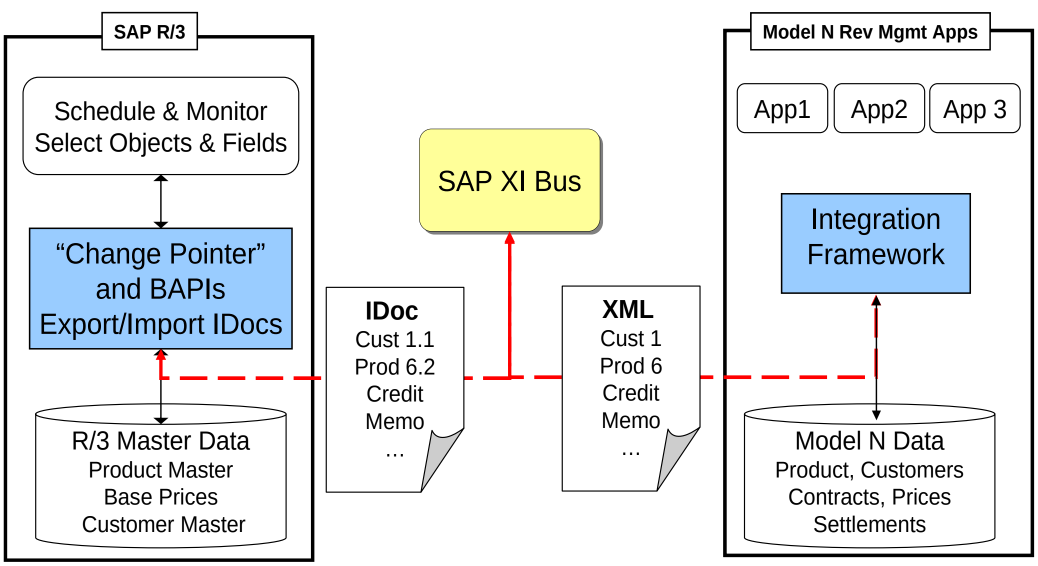
- Master data from SAP — customer/address records, product/material records, base price lists
- Transactional data from SAP — direct/indirect sales lines, rebate payments, financial adjustments
- Settlement data to SAP — rebate payment claims, admin fee claims, chargeback credit memos, and sometimes rebate accruals

In a standard SAP environment, Model N generally transmits XML and iDoc files via SAP's XI bus. Model N has developed and makes available pre-defined XI maps. These maps can be loaded into the XI system, where they can be easily modified to match the customer's SAP implementation.

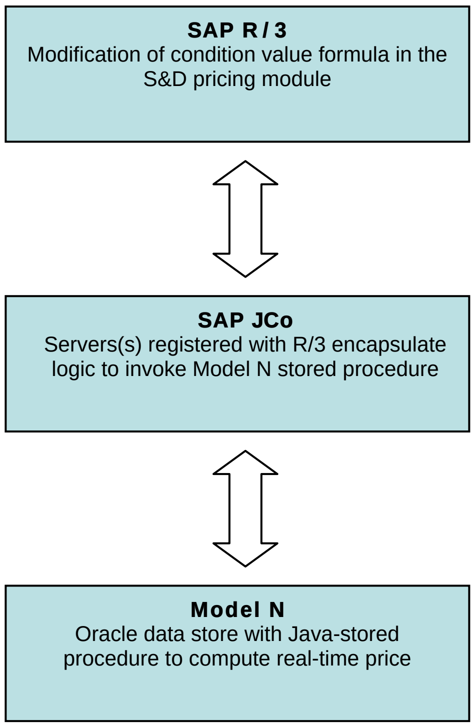
A sample map between SAP's product iDoc and Model N's XML is portrayed below.



The diagram below depicts a high-level picture of the data flow between SAP's R/3 system and the Model N Revenue Management Suite using an EAI bus (XI). The XI bus can be replaced with an alternate EAI system, or with a direct file channel that relies on maps within the Model N RM integration server.



Apart from the asynchronous data integration described above, integration between RM and ERP systems also requires a real-time component. The Model N **real-time connector** is used by the ERP system to resolve “ambiguous” prices against multiple overlapping contracts, membership records, and customer commitments to various tiers and programs. To support critical transaction processing and security requirements, the Model N price resolution engine is deployed as a stored procedure in the RDBMS and accessed in real time by ERP pricing procedures. When orders are placed in the R/3 system, real-time prices are resolved against the Model N price engine.

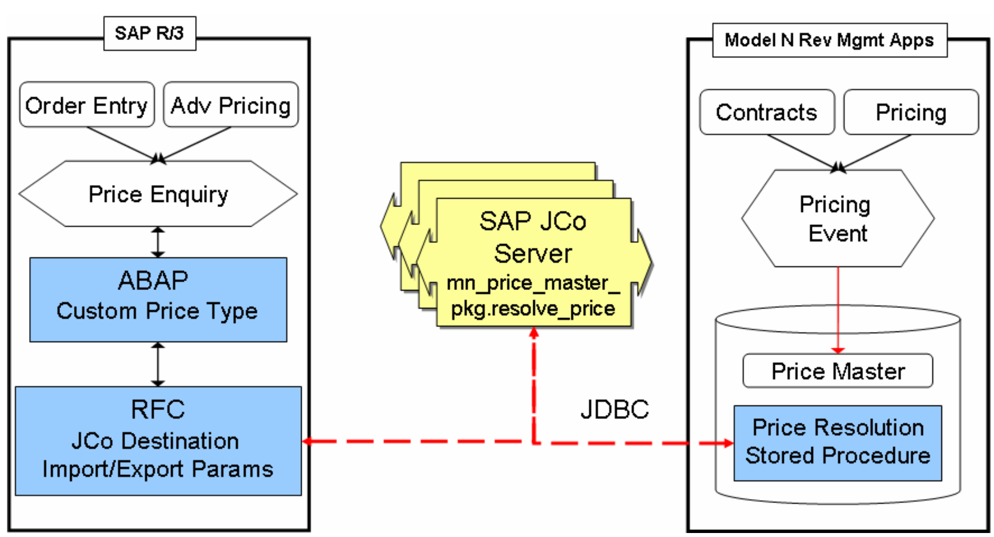


The advanced pricing procedures of R/3 (in the S&D module) are enhanced to query the Model N data base for the exact net price for a particular customer and product on a particular date. The R/3 price condition (PR00) is enhanced with a custom condition value formula, as well as appropriate import/export parameters to enable the remote Model N price procedure.

The preferred method of implementing the call-out is via an RFC to a SAP JCo server. JCo is the SAP standard for communicating between R/3 and external Java systems.

On the backend of the call, Model N provides a "Java-stored procedure package” installed in the database. It computes and returns the price lookup (price, reason code, etc.) to the R/3 system.

The results returned by the Model N real-time price connector are usually used as a base price override to the pricing detail line condition in the relevant SAP services (e.g. sales price procedure in the S&D module). Thereafter, all further price processing in the R/3 system proceeds normally, invoking other price conditions such as duty, transportation, sales order footer, pricing, etc.
The diagram below depicts the real-time RFC callout from the ABAP price procedure via a JCo server to the stored procedure in the Model N RM database.



Conclusion

As noted in the introduction, life sciences manufacturers are successfully deploying RM solutions in conjunction with their ERP/legacy systems. This paper showcases the need for a rigorous and formal integration capability between RM and ERP systems while outlining the implications of this integrated picture on the IT enterprise landscape. We have looked at some principle solution points in detail, and reviewed specific aspects of Model N's investment in the product and experience in the field integrating with R/3.

Any RM deployment in the ERP landscape has unique and differing requirements and there is no single cookie-cutter approach. Successful deployments are based on executive buy-in and support, investments in quality data and processes, and the work of specialized teams of data analysts and integration engineers who focus entirely on the data landscape and integration services throughout the project lifecycle. Ultimately however, mandating an architectural approach that supports a high degree of configuration and flexibility is a fundamental ingredient of success.

Revenue Management is a strategic requirement for businesses in today's competitive and regulatory climate. Going forward, trends only point to a continued need for flexibility in both the core systems and processes built to optimize revenues and profitability. For business executives, it is no longer a question of 'When?' This paper provides perspectives and experiences supporting the dialog that IT managers and executives routinely undertake as they seek to answer the question: 'How?'

About Model N

Model N is the leader in Revenue Management solutions, offering an integrated suite of applications for pricing, contracts, compliance, rebates, fees, and chargebacks optimized for the industry practices of life sciences companies, with the only integrated government pricing and Medicaid claims processing capabilities in the industry. Enabling the creation of a seamless, end-to-end process from pricing through settlements payment, Model N's uniquely integrated approach eliminates revenue leakage and delivers the visibility and controls needed to avoid the risks of non-compliance to government pricing and Sarbanes-Oxley regulations. Customers include Actelion Biopharmaceuticals; Axcan Pharma; Boston Scientific; Bristol-Myers Squibb; CoTherix, Inc.; Gilead Sciences; MedImmune, Inc.; Medtronic, Inc.; Novo Nordisk, Inc.; Ortho-Clinical Diagnostics, a Johnson & Johnson company; and Pfizer, Inc. Model N is located in Redwood Shores, California. For more information on Model N Revenue Management solutions, please visit www.modeln.com or contact us at 866-266-3356.